

InstructGPT: Reinforcement Learning from Human Feedback with PPO (& DPO!)

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Introduction & Motivation

- **Alignment Problem:** Standard LLMs trained exclusively on next-token prediction objective usually optimize for text completion accuracy over being **helpful, honest, and harmless** (three H's!)
- **InstructGPT (Ouyang et al., 2022) introduced RLHF:**
 - 1) **Supervised Fine-Tuning (SFT)** on instruction demonstrations.
 - 2) **Reward Modeling (RM)** via binary human preference pairs.
 - 3) **Proximal Policy Optimization (PPO)** to maximize expected rewards.
- **Key Result:** Alignment drastically alters model scaling efficiency, proving that outputs from a small **1.3B** parameter aligned model are consistently preferred over a massive **175B** parameter unaligned base model.
- **Project Objective:** To reproduce the alignment pipeline of InstructGPT utilizing a smaller, efficient model (distilgpt2, 82M parameters).
- **Hypothesis:** Post-training alignment frameworks (RLHF with PPO and DPO) will enhance alignment even with a smaller model.

Methodology

- Stage 1: Supervised Fine-Tuning (SFT)**
- Goal: Shift the model's output from generic unstructured text prediction toward instruction-response sequences.
 - DistilGPT2 (82M parameters); Stanford Alpaca (52k instruction pairs)
 - Fine-tuned with SFTTrainer; AdamW over a 500-step trajectory.
- Stage 2: Reward Modeling (RM)**
- Goal: Train a proxy evaluator capable of scoring response quality according to human baseline preferences.
 - Sequence Classifier (DistilGPT2 base); Anthropic HH-RLHF (5k samples)
 - Loss: Binary Ranking Log-Odds Loss
- Stage 3: Proximal Policy Optimization (PPO)**
- SFT model acts as agent, generating responses that are graded by the trained reward model.
 - KL-divergence penalty (target 0.1) to prevent reward hacking and extreme divergence from original model
- Our Addition: Direct Preference Optimization (DPO)**
- Dataset: UltraFeedback Binarized
 - Reward-free alignment optimizing policy directly on preference pairs.
 - Advantage: Eliminates PPO training instability, the need for a separate reward model, better computational efficiency than PPO

Results



Model Performance Comparison

SFT beats PPO: 20.00%
SFT beats DPO: 30.00%
PPO beats SFT: 80.00%
PPO beats DPO: 70.00%
DPO beats SFT: 70.00%
DPO beats PPO: 30.00%

Win Rates! SFT vs RLHF (with PPO) vs RLHF (with DPO)

Response Comparison

Prompt	SFT (Baseline)	RLHF with PPO	RLHF with DPO
Give a tip for staying healthy.	Stay healthy, and stay healthy.	Being healthy is an important aspect of life.	Keep an eye on your activity and exercise plan.
Explain what a solar eclipse is.	The solar eclipse is the largest solar event in history.	A solar eclipse is a celestial object (lit by the Sun).	The eclipse is a major event in the solar system.
Write a short greeting for a new neighbor.	Hello, your new neighbor is a new neighbor.	I love you very much.	Hello and welcome to my new neighbor. Thank you very much for all the love and support you have given me. I am so excited to be here.
Explain what machine learning is.	Machine learning is a powerful and effective way to teach students how to code.	Machine learning is used to identify and retrieve information about objects and behaviors in a variety of fields.	Machine learning is an approach in which we measure the number of neurons, which are thought to be connected to the neuron in the brain.

Conclusion

- Alignment via PPO and DPO helped shift the DistilGPT2 output distribution toward more helpful, coherent responses (RLHF works!)
- Scalability: Post-training alignment was effective even with a small model, demonstrating that RLHF isn't exclusive to multi-billion parameter architectures.
- DPO, in theory, is more stable compared to PPO. However, RLHF with PPO still yielded better performance in our experiments

Future Work

- Architectural Scaling: Apply the RLHF pipeline to larger open-source models using Parameter-Efficient Fine-Tuning (PEFT) techniques like LoRA or QLoRA
- RLAI: Try Reinforcement Learning from AI Feedback, where a larger "judge" model generates the preference labels, removing the bottleneck of human-annotated datasets like the Anthropic HH-RLHF we used

References

1. Ouyang et al. (2022): Training language models to follow instructions with human feedback
2. Rafailov et al. (2023): Direct Preference Optimization: Your Language Model is Secretly a Reward Model